

LOURENZ O. BALIBER, MSc

Researcher | Soft Matter | Physics | Data Scientist

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Physics postgraduate and data scientist with expertise in soft matter and polymer physics, end-to-end machine learning pipelines, predictive modeling, and data-driven application deployment. Proficient in Python, R, and MATLAB with hands-on experience in data analysis, computer vision, and scientific computing. Currently a data science intern at OpsWerks, building and evaluating LLM applications (retrieval-augmented generation and agents), with a track record of collaborating with international research teams.

TECHNICAL SKILLS

- Programming & Data Tools: **Python, R, MATLAB, Mathematica, C++, SQL, Git, Linux/shell**
- Data Science & ML: **Machine learning, forecasting, statistical modeling, NLP, computer vision**
- LLM & Generative AI: **RAG, LLM agents, prompt engineering, guardrails, LLM evaluation**
- Deployment & Web: **Streamlit, Flask, React, REST APIs, Docker, MongoDB, JavaScript**
- Scientific Computing: **Molecular dynamics, physics simulation, microrheology, particle tracking**
- Other: **Tableau, LaTeX, scientific writing, CAD modeling, technical presentations**

EXPERIENCE

Data Science Intern

Aug 2026 – Present

OpsWerks, Cebu City, Philippines

- Selected for a **12-week full-time data science program** spanning foundations, intermediate machine learning, and a mentor-approved capstone project.
- Analyzed **997 e-commerce support conversations** (sentiment classification at 87.5% accuracy, NMF topic modeling, embedding clustering) and delivered a 7-page Streamlit dashboard in Docker with a briefing for managers.
- Built a **RAG chatbot** over 9,000+ Philippine food product records and labeling regulations (LangChain, Chroma, SQL on DuckDB) with cited answers; retrieval hit rate 0.95 at k=10 on a labeled question set.
- Built a **LangGraph data-quality agent** (capstone) that reviews records through tool calls and drafts corrections; passed 24 of 24 labeled review tasks, alongside a category classifier at macro-F1 0.71 (zero-shot LLM: 0.50).
- Maintaining the required **80% average** across all technical assessments; completed **26 DataCamp courses** (Python, machine learning, NLP, time series, LLMs, LangChain, Git, Docker).

Exchange Student Researcher (Physical Chemistry)

2025 – 2026

Tokyo University of Marine Science and Technology

- Processed and analyzed **2,000+ fluorescence microscopy image frames** using custom Mathematica pipelines, reducing manual analysis time by **~60%**.
- Developed image analysis workflows correlating structural heterogeneity with particle dynamics from passive Brownian motion tracking across **40+ samples**.
- Applied statistical modeling to characterize phase-separated biopolymer network formation; results and data analysis contributed to a **peer-reviewed publication**.

Graduate Researcher – MSc Thesis

2024 – 2026

Medical Biophysics Group, University of San Carlos

- Performed quantitative analysis of fluorescence microscopy and passive particle tracking on mixed carrageenan hydrogel systems.
- Implemented data processing pipelines in Python/MATLAB for MSD, viscoelastic moduli extraction, and gel network characterization.
- Collaborated with international researchers on soft matter physics and data analysis methodology.

Research Assistant

Aug – Nov 2023

DOST PCIEERD NICER Project — 3D-BTM: Bio-fabrication of Bone, Muscle & Pancreatic Tissue Models for Printable and Injectable Scaffolds

- Supported data collection and analysis for biomimetic tissue engineering experiments, processing **15+ experimental datasets**.
- Assisted in processing and interpreting biomechanical measurement datasets; contributed to a **government-funded national research project**.

Research Intern

Jun – Jul 2023

Medical Biophysics Group, University of San Carlos

- Diagnosed optical tweezers and microrheology experiments on complex fluids.
- Processed experimental data to quantify local viscoelastic properties of biological samples, achieving reproducibility within **±3% measurement error**.

EDUCATION

MSc – Physics

2024 – 2026

University of San Carlos, Cebu City, Philippines

- Thesis: "Gelation Mechanism and Network Formation of FITC-Labeled Mixed Kappa/Lambda Carrageenan Gels"
- DOST ASTHRDP Scholar | GWA: 1.19

BSc – Applied Physics

2020 – 2024

University of San Carlos, Cebu City, Philippines

- Thesis: "Characterization of Fibrin-Carrageenan Mixtures as a Hydrogel Model Using Passive Particle Tracking and AFM for Skeletal Muscle Tissue Engineering"
- DOST JLSS Scholar (2022–2024) | Ironwood Scholar (2020–2024) | Dean's Lister (2020–2022)

PUBLICATIONS

- Baliber, L. et al. (2026). Brownian motion in a phase-separated biopolymer network. *Proceedings of the 44th Samahang Pisika ng Pilipinas Physics Conference*, SPP-2026-PA-01.
- Baliber, L., et al. (2025). Surface roughness and fractal dimension characterization of the topography in pristine and defected nanostructure arrays. *Proceedings of the Samahang Pisika ng Pilipinas*, 43, SPP-2025-2G-02.
- Baliber, L., et al. (2024). Passive microrheology in carrageenan–fibrin mixtures using video microscopy. *Proceedings of the Samahang Pisika ng Pilipinas*, 42, SPP-2024-1E-05.

PROJECTS

Coarse-Grained Polymer Simulation — Molecular Dynamics

2026

LAMMPS · Python · OVITO | github.com/loubaliber/cg-polymer-md

- Built coarse-grained molecular dynamics simulations of polymer chains in **LAMMPS**, with Python-driven setup, trajectory post-processing, and OVITO visualization of chain conformation.

Signal Translator — Audio Fingerprinting Desktop App

2026

Python · FFT · MFCC | github.com/loubaliber/signal-translator

- Fingerprints short vocal signals from the microphone with **FFT and MFCC audio features** and matches new input against a library of user-taught signals, so custom "words" can be learned and recognized.

Hands Detection Music — Real-Time Computer Vision Instrument

2026

Python · OpenCV | github.com/loubaliber/hands-detection-music

- Real-time webcam hand tracking mapped to live audio synthesis: the left hand selects the chord and the right hand controls how it is played, running end-to-end on a standard laptop camera feed.

Climate Delhi — Time-Series EDA & Regression

2026

Python · Pandas · Scikit-learn | github.com/loubaliber/climate-delhi

- EDA and regression notebook predicting daily mean temperature in Delhi from the **2013–2017 Daily Climate Time Series** dataset, covering feature engineering, seasonality, and model evaluation.

Satellite Launch & Orbit Simulator — Python Orbital Mechanics Model

2025

Python · RK4 · Matplotlib

- Built a 2D simulation of a satellite's trajectory from launch to LEO insertion using **custom RK4 integration** under gravity and atmospheric drag with fuel mass loss; validated against analytical orbital mechanics solutions.

Stroke Detection System — End-to-End ML Pipeline

2021

Flask · Streamlit · Scikit-learn | github.com/loubaliber/Stroke-Dataset-Prediction

- First-year undergraduate project: built a stroke risk classifier from EDA to web deployment on **5,000+ patient records**, served as a Flask + Streamlit app with real-time risk prediction and feature importance.

Additional Tools

2023 – 2026

- **Balibot** — Discord bot built with discord.py serving **200+ users across 5+ servers**; automated moderation, slash commands, API integrations, **96.3% uptime** over 6 months.
- **latex-hub.nvim** — Neovim plugin (Lua) with a project switcher and status dashboard for LaTeX projects.
- **Gmail Scraper** — Python/IMAP utility that archives a Gmail mailbox and its attachments to text files.

HONORS & AWARDS

- ICBBS 2026 (International Conference on Biophysics and Biomedical Sciences), BEXCO, Busan, South Korea – Poster Presenter (accepted, Nov 2026)
- 25th SPVM National Physics Conference – Poster Presenter (2023)
- 33rd Annual Fr. Heinrich Schoenig Biosymposium – Poster Presenter (2024)
- ODU REYES Python & Subatomic Physics Mentorship (2021)
- Top 3 Scholar, USC DOST Thesis Grants (2024)
- 44th SPP Conference – Poster Presenter (2026)

AFFILIATIONS

- Creative Writer - The Josenian Premier (2019 - 2020)
- **Administrative Positions:** Internal Affairs Councilor, Carolinian Physics Society (2021–2022); President, USC Physics and Astronomy Society (2022–2023); Councilor, School of Arts and Sciences Council (2022–2023); Physics Representative, DOST ASTHRDP Officer (2024–2025)
- **Co-organized Conferences:** 2nd and 3rd USC-VSU Biophysics Symposium (2023, 2025); 8th MBG Biophysics Symposium (2024); 8th International and 16th Annual National Scientific Meeting of the Philippine Society for Cell Biology (2025); 18th International Hydrocolloids Conference (2026)